

Sai Mun Lee

SENIOR AI ENGINEER | LLM & AGENTIC SYSTEMS

☎ (+65) 9336 4288 | ✉ saimun92@icloud.com | 🏠 saimunlee.com | 🌐 lee-sai-mun-8a12726

Summary

Senior software engineer with 8+ years shipping production AI/ML products end to end in enterprise SaaS — from architecture and API design to serving, reliability, latency and cost. Built and operates SAP's multi-tenant foundation-model serving platform (10+ enterprise tenants, 8B+ recommendations served in 2025), owning inference orchestration, tenant isolation and data governance. Partners closely with product management as a co-owner, leads 15-person teams through zero-to-one, high-ambiguity initiatives, and mentors engineers. Builds hands-on with LLM and agentic stacks — agent frameworks, tool/function calling, RAG and prompt engineering (prototyping) — and works fluently with modern gen-AI developer tooling.

Skills

LLM & Agentic AI	Agent Frameworks, Tool/Function Calling, Multi-Agent Orchestration (prototyping)
RAG & LLM APIs	RAG, Prompt Engineering, Vector Stores (pgvector), LLM APIs (Claude, OpenAI, Gemini)
ML & Serving	Foundation-Model Serving, Model Inference & Optimization (latency, cost), Recommender Systems, PyTorch
Languages	Python, Go, TypeScript, JavaScript, Bash
Backend & APIs	REST API Design, Microservices, FastAPI, Flask, Django, PostgreSQL, HANA DB, Elasticsearch
Infrastructure	Kubernetes, Docker, Cloud Foundry, Jenkins, Terraform
Cloud	AWS (Solutions Architect Pro, in progress), GCP (Associate Cloud Engineer)
Security	Multi-tenant Isolation, Encryption (at-rest & in-transit)
Practices	System Design, Distributed Systems, ML Platform Architecture, Cross-functional Leadership, Code Review, TDD
Spoken	English, Chinese

Experience

SAP Asia

Singapore

SENIOR SOFTWARE ENGINEER | CO-PRODUCT OWNER

Dec 2020 - Present

- Architected and shipped SAP's Tabular AI platform end to end — a model-agnostic, highly scalable multi-tenant serving layer for the RPT-1 foundation model on Kubernetes (SAP AI Core), targeting 10+ enterprise tenants at 100K users each.
- Owned the full lifecycle of AI features: technical design, API contracts, model selection and inference orchestration, tenant isolation, production deployment and monitoring — ensuring reliability, latency and cost efficiency at scale.
- Optimized the model inference serving path to improve latency, throughput and cost under concurrent multi-tenant workloads.
- Partnered closely with product management as co-owner — and with design, infrastructure and security teams — to translate business needs into technical architecture and drive decisions under ambiguity.
- Implemented enterprise security — encryption at rest and in transit, and tenant data isolation — for employee data on SAP SuccessFactors.
- Led a 15-person cross-functional team delivering Personalized Recommendations, a recommender engine that generated 8B+ recommendations in 2025 for enterprise customers, and architected People Like Me, a service that aggregates customer data into training data for Personalized Recommendations and exposes its own REST API for consuming recommendations.
- Set engineering best practices and mentored engineers through code review and pair programming; established CI/CD (Jenkins) with unit, integration and e2e coverage and instrumented the platform with Dynatrace APM and Kibana (correlation-ID tracing), reducing mean-time-to-resolution for production incidents.

TAU Express

Singapore

LEAD SOFTWARE ENGINEER

Apr. 2020 - Dec. 2020

- Led development of a knowledge-management and document-retrieval system enabling analysts to retrieve and correlate insights across cybersecurity documents via graph relationships (Neo4J) and full-text search (ElasticSearch) — retrieval foundations directly relevant to RAG.
- Owned end-to-end delivery: gathered requirements directly from users, designed the system architecture, and facilitated communication between users and the development team.
- Tech: React, Django, Docker, Neo4J, ElasticSearch, Cytoscape.js

SPIRIT Research Center (NTU)

Singapore

RESEARCH ENGINEER

Feb. 2018 - Mar. 2020

- Developed the Intelligent Case Retrieval System (ICRS) for Singapore's Smart Nation initiative — a full-stack document-retrieval application (Django, Vue.js) for efficient retrieval of legal documents.
- Built web crawlers (Selenium, BeautifulSoup) to index Chinese news articles and social media content for document-retrieval pipelines.

Certificates

- Dec 2021 **AWS Certified Solutions Architect – Associate**, Amazon Web Services (AWS)
- Feb 2022 **GCP Associate Cloud Engineer**, Google Cloud Platform (GCP)
- May 2023 **Hashicorp Certified: Terraform Associate (003)**, Hashicorp
- AWS Certified Solutions Architect - Professional**, Amazon Web Services (AWS) *In Progress*

Education

Nanyang Technological University (NTU) *Singapore*
B.S. IN COMPUTER SCIENCE (HONOURS) — MACHINE LEARNING & DATA SCIENCE SPECIALISATION 2013 - 2017